

AI/ML undergraduate with hands-on experience in Generative AI, NLP, LLM fine-tuning, and RAG systems. Built multiple end-to-end AI applications using PyTorch, LangChain, and vector databases, with interests in scalable ML systems and AI security.

EDUCATION

Examination	Institute	Year	CGPA/%
Graduation (B.Tech, AI & ML)	Sasi Institute of Technology and Engineering	2023-2027 (Expected)	8.5/10.0
Intermediate/+2	Sasi Junior College, Tadepalligudem	2021-2023	97.6%
Matriculation	Bhashyam Public School, Tadepalligudem	2021	100%

RELEVANT COURSEWORK

- Machine Learning • Deep Learning • Data Structures & Algorithms • Operating Systems • DBMS • Computer Networks

TECHNICAL SKILLS

- AI/ML Core:** Machine Learning, Deep Learning, Generative AI, LLM Fine-tuning (Llama 3, Mistral), Transformers, RAG, Computer Vision, NLP.
- Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Hugging Face, LangChain, OpenAI API, Pinecone.
- Programming:** Python, Java, SQL, Data Structures & Algorithms (DSA).
- Deployment & Tools:** Docker, Kubernetes, AWS SageMaker, Flask/FastAPI, REST APIs, Gradio, Git, CI/CD, MLOps, Linux.

PROFESSIONAL EXPERIENCE

- GenAI & Multi-Agent Orchestration Intern | PurpleLane**May 2026 - Jul 2026
 - Worked on **Generative AI systems**, architecting **multi-agent orchestration** frameworks to automate complex, multi-step workflows.
 - Applied **deep learning** techniques to meaningfully improve model performance and pipeline response time.
 - Collaborated on integrating **LLM-based agents** into orchestration systems for end-to-end task automation.

PROJECTS

- PennyWise-AI | Generative AI & RAG** <https://github.com/Harshitha-Madhu/PennyWiseAI>
 - Engineered** a financial intelligence platform using **Retrieval-Augmented Generation (RAG)** and **Gemini** for zero-shot classification.
 - Integrated a **Pinecone vector database** for semantic search, enabling natural-language queries over unstructured financial data.
 - Automated** categorization of spending patterns, eliminating manual data entry requirements by 100%.
- HealthHarbourAI | Multimodal Medical Diagnostic System** <https://github.com/Harshitha-Madhu/HealthHarbourAI>
 - Architected** a multimodal pipeline using **Gemini 1.5 Pro**, fusing medical imaging analysis with evidence-based diagnostic suggestions.
 - Reduced** inference latency by **30%** through optimization techniques, powering real-time interaction via a **Gradio** frontend.
 - Transformed unstructured clinical inputs into structured, actionable medical insights to support diagnostic decision-making.
- Production-Grade ML Simulation | MLOps & System Design** <https://github.com/Harshitha-Madhu/Production-Grade-Simulation>
 - Built** a robust ML lifecycle simulation using **Docker**, bridging the gap between research models and scalable production infrastructure.
 - Implemented **CI/CD** principles and containerization, significantly reducing deployment time across distributed environments.

- Engineered for observability and scalability, aligning with enterprise-grade standards for secure, next-gen AI systems.
- **Revenue-Fraud Detection Engine | Predictive Analytics** <https://github.com/Harshitha-Madhu/Revenue-Fraud>
 - **Engineered** a high-performance detection engine using **XGBoost** and **Random Forest** to flag financial anomalies in real time.
 - Conducted advanced feature engineering to reduce false positives by **20%** in a simulated production environment.
 - Prioritized low-latency scoring to trigger instantaneous security alerts across high-velocity transaction streams.
- **AI-Document-Anonymizer | NLP & OCR** <https://github.com/Harshitha-Madhu/AI-Document-Anonymizer>
 - **Developed** an automated pipeline for **OCR**, text classification, and abstractive summarization using **Hugging Face Transformers**.
 - Delivered a functional **Proof of Concept (PoC)** that substantially reduced manual document review time.
- **FingerDiab | Non-Invasive Diabetes Risk Prediction System** <https://github.com/Harshitha-Madhu/FingerDiab>
 - **Built** a biometric ML system using **OpenCV**, **Flask**, and **Random Forest**, achieving strong classification accuracy on fingerprint and health-parameter data for real-time diabetes risk prediction.
- **Synthetic Financial Data Generation Engine** <https://github.com/Harshitha-Madhu/Finance-TimeSeries-GAN-Engine>
 - **Designed** an efficient **MLP-based 1D GAN** architecture, reducing training compute overhead while enabling scalable, privacy-preserving synthetic financial data generation.

CERTIFICATIONS

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| ● AWS Generative AI: Specialization in Diffusion Models | <i>Feb 2026</i> |
| ● Oracle AI-Vector Certification: Focus on Vector Databases and AI integration | <i>Oct 2025</i> |
| ● IBM SkillsBuild: Build an AI Agent | <i>Mar 2026</i> |
| ● Cisco: Introduction to Modern AI (Course & Badge) | <i>Feb 2026</i> |

HONORS & LEADERSHIP

- 2-Star Coder on CodeChef.
- Ethical Hacking Recognition: Certified Enthusiast by Supraja Technologies.
- Open-Source Contributor: Actively maintaining AI repositories with documented deployment guides.